Preparing CLASP-seq libraries

From sequence lists or plasmids to qualified receptor and HLA-peptide cell libraries.

Working document. Red **TO COMPLETE** fields identify missing or conflicting author inputs. Complete the fields affecting a procedure before using it at the bench. Recorded settings are not a claim of experimental optimization.

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01 Choose the starting material

The assay requires separate TCR-expressing Jurkat and HLA-peptide-expressing K562 libraries, with records that connect each construct to its biological identity. Skip a preparation stage only when its qualified materials and records are available. [S1 p.1]

Recorded settings below come from the author-supplied source synthesis. They describe the documented workflow; they have not been established as universal optima. Red completion fields remain attached to the steps they affect.

Available material	Work needed
Sequence lists	Build and verify plasmid libraries, then generate and qualify cell libraries.
Verified plasmid libraries	Generate and qualify cell libraries using the supported packaging/transduction method.
Qualified cell libraries	Review identity, expression, viability, representation and reference records; proceed to the assay.

TO COMPLETE - C01
Supply exact cell derivatives and sources, engineering/authentication history, full culture media and conditions, cell-quality checks, accessible vector sequences and validated library-qualification criteria.

TO COMPLETE - C14
Supply authors/contact, released preparation version, material access routes and reuse/distribution terms. Local CZ/MZ names alone are not sufficient for another laboratory to obtain the materials.

02 Define and verify every construct

The overview records the following component order. These are schematics, not complete cloning maps. The documented HLA-peptide display architecture is class I style. [S1 p.1]

Library	Recorded architecture
TCR	EF1a:[TSO][TCRa/b][CS1].IRES.PURO
HLA-peptide	EF1a:[TSO][GFP][T2A][Sigpep][PEP][B2M][HLA][CS1]

1. Prepare the intended TCR and HLA-peptide identity list. Preserve the linked α/β receptor identity, actual HLA allele, peptide and any engineered changes.
2. Provide full DNA and translated sequences, orientation, signal peptides/linkers and all primer-binding coordinates for each supported construct.
3. Assign stable construct and library identifiers. Link sequence verification, expected amplicon sizes and observed library representation to those identifiers.
4. Retain missing, unexpected and ambiguous library members. A correct-size band does not establish sequence identity or composition.

TO COMPLETE - C01
Reconcile MZ-1/MZ-01 and the supported lentiviral backbone. Supply annotated HLA receiver maps, including allele-specific substitutions and restriction sites. Repair and resequence flagged HLA-B*51:01 stocks CZ4528/CZ4528R and HLA-C*01:02 stock CZ4532 before use; the source identifies them as requiring recloning. [S3 p.1]

Optional upstream TCRAFT route

TCRAFT may supply a TCR-library construction route. Its notes describe oligo design, pooled cloning and transfer to a lentiviral backbone, but contain an unfinished local adaptation, an unresolved digest and conflicting vector labels. Use a separately versioned, verified construction SOP if offering this route. It is not required when a compatible, qualified TCR library is already available. [S2 pp.2-3,12-13]

TO COMPLETE - C02
For the offered TCR-construction route, supply oligo-design rules, exact backbone, full cloning/digest/transfer recipes and sequence-level QC. Close the TCRAFT adaptation conflicts if that route is included. Its representation-PCR primers do not fill the missing CLASP-seq target-recovery PCR recipe.

03 Build the HLA-peptide plasmid library

The recorded pooled-library route is oligo amplification, backbone digestion and purification, assembly, bacterial transformation, plasmid preparation and qualification. Preserve the exact pool and vector identity through each stage. [S3 pp.2-4]

Oligo-pool amplification

Component	Recorded volume
Water	6 µL
CZP22 forward primer	0.75 µL
CZ5 reverse primer	0.75 µL
KOD 2X buffer	15 µL
dNTPs	6 µL
KOD enzyme	0.5 µL
Oligo-pool template	1 µL
Total, calculated from recorded volumes	30 µL

The recorded program is 98 °C for 3 min; then 12 cycles of 98 °C for 15 s, 50 °C for 15 s and 68 °C for 20 s. The example product is 81 bp, followed by MinElute cleanup and Qubit measurement. A final extension/hold is not specified. [S3 pp.2-3]

TO COMPLETE - C02
Complete the exact KOD system, primer/dNTP stocks, pool/template identity and concentration, full program and cleanup. The source alternates between approximately 2.5 ng/µL and 5 ng/µL template preparations; a separate registry entry refers to CZP24 and 15 cycles. Reconcile these with the chosen pool rather than combining them. [S3 pp.2-3; S10 p.1]

Backbone digestion and purification

The recorded digest uses 2 µL 'CS buffer', 14 µL water, 2 µL plasmid containing 1 µg, 1 µL BamHI and 1 µL AfeI. Incubate at 37 °C for 3 h, gel-purify and elute in 20 µL. This recipe is conditional on the exact vector map. [S3 pp.1,3]

TO COMPLETE - C02
Identify the buffer/product and enzyme specifications, gel/extraction procedure and verified digest for each backbone. HLA variants have different BamHI/AfeI site counts; the same digest is not established for every listed plasmid.

Pooled assembly and transformation

TO COMPLETE - C02
Resolve vector mass/concentration and intended insert/vector relationship before assembly. Supply competent cells, dialysis/electroporation/recovery conditions, plating and expansion strategy, plasmid preparation and qualification method. The separate small-sequence cold-fusion branch is not established as interchangeable for pooled libraries.

The assembly note lists 1 µL vector, 2 µL peptide insert preparation, 1 µL water and 4 µL 2X HiFi mix; incubate for 20 min at 50 °C, then dialyze and electroporate. The note also lists insert concentration 0.5 ng/µL and an unexplained adjacent '1ug plasmid' annotation. [S3 p.4]

Plasmid checkpoint
Verify sequences, architecture and representation before cell generation. Retain unexpected/missing members and abundance measurements. The source proposes PCR/restriction checks and amplicon sequencing, but describes its PEP distribution assay as untried; provide a qualified method and expected outputs. Optional clone deconvolution requires its own complete SOP.

04 Generate and characterize the vector preparation

The source uses HEK293FT packaging cells and records the schedule below. Follow the receiving laboratory's approved vector and cell-handling procedures. Complete material identities, medium formulations and transfection recipes before using the schedule. [S4 p.1]

Day	Recorded step
-1	Plate 1.02 × 10^6 log-phase HEK293FT cells per 6-well.
0	At 90% confluence, transfect with the Lipofectamine/DNA cocktail; the source specifies afternoon.
1	Replace with collection medium: HEK293FT medium, 25 mM HEPES and 'Viral boost' at 30 µL/15 mL.
2	Collect 2 mL and replace collection medium.
3	Collect 2 mL, combine with the preceding harvest, filter and characterize.

TO COMPLETE - C02
Supply transfer/packaging plasmid identities and maps, DNA amounts/ratios, Lipofectamine formulation and dose, culture and transfection volumes, full media, 'Viral boost' identity, filtration, harvest handling/storage and functional-titer method in the actual recipient cells.

The notes use 'Gostix p24' and 'IFU/mL' without a defined calibration between them. Record what the assay measured and its units. Do not convert an unspecified p24 measurement into a functional infectious titer or treat nominal MOI as verified delivery. [S4 p.1]

05 Transduce and recover the recipient libraries

The recorded spinfection sequence is given below. Dose branches are local source examples, not a demonstrated optimum for each cell derivative or library. Keep dose conditions separate through qualification. [S5 pp.1-2]

1. Use healthy, log-phase recipient cells in unspent medium. The source typically uses 300,000-1,000,000 cells and pellets at 300 × g for 3 min.

2. Prepare separate dose conditions. The notes describe nominal MOI 1 and MOI 3 branches, with polybrene targeted at 8 µg/mL final. Calculate from a verified stock, actual final mixture volume and a defined functional-titer basis.
3. Incubate at 37 °C for 30-60 min; the source flicks every 15 min. Distribute approximately 200 µL/well into the specified U-bottom, non-tissue-culture-treated plate.
4. Centrifuge in the approved balanced/contained setup at 1,200 × g, 32 °C for 2 h.
5. Recover at 37 °C for 6 h. Remove supernatant without disturbing the pellet, resuspend in fresh recipient-cell medium and recombine split wells from the same condition. The source example is 1,000,000 cells in 2 mL in a 6-well.
6. Record delivered dose, expression/infection measurement, viability and recovery. Retain each condition's identity through selection.

TO COMPLETE - C02
Complete polybrene product/stock, final-volume calculation, functional-dose basis, recipient-specific medium and recovery conditions, plate/rotor configuration, mixing/handling and progression criteria. The source's '4 µL into 500 µL' example is not usable without its stock definition.

TCR/Jurkat selection

The more specific source schedule starts puromycin 72 h after spinfection at 0.4 µg/mL, with fresh selection medium daily for four full days. It then uses Miltenyi live/dead magnetic selection, recovery without puromycin and expansion. The spinfection sheet also says 48-72 h elsewhere. [S4 p.1; S5 p.2]

TO COMPLETE - C02
Select and validate the puromycin start time, product/stock, full selection/recovery procedure and magnetic-selection kit/method. Do not silently resolve 72 h versus 48-72 h by treating the schedules as equivalent.

HLA-peptide/K562 selection

The source sorts GFP-positive cells on day 3-4. Record the actual day, complete gate/control setup, fraction positive, recovery and representation. GFP positivity alone does not establish each displayed construct's identity or surface abundance. [S4 p.1]

TO COMPLETE - C01
Supply construct-expression qualification, including measurement reagents/controls and acceptable results, for each library. Define when a selected population is ready for banking and assay use.

06 Preserve representation and bank stocks

The source favors the lowest infection condition that preserves sufficient representation and tends toward one library member per cell. A nominal MOI or positive marker does not establish one construct per cell. The suggestion that multiple integrations may be resolved computationally is not a demonstrated correction method. [S4 pp.1-2; S5 pp.1-2]

1. Measure and retain representation at the plasmid and qualified cell-library stages. Preserve missing/unexpected members and the observed abundance distribution.
2. Record bottlenecks, viable recovery, selection losses, expansion and freeze/thaw history. Expanded cell counts are not a direct count of independent founder events.

3. Keep unqualified or depleted libraries distinguishable from usable assay material. Expansion cannot restore missing members.

The cell-generation notes call for 3-4 frozen stock vials before experiments. The K562 freeze sheet compares 5% with 10% final DMSO and selects 5% for the author's practice. That local selection does not establish the best condition for every derivative. [S4 p.2; S6 pp.1-2]

Recorded K562 freezing branch

1. Prepare FBS with 10% DMSO as a 2X mixture. Prepare cells in their normal medium at 1,000,000 cells per 0.5 mL.
2. Chill the cells, mixture and tubes on ice for 10 min.
3. Add an equal volume of the 2X mixture gradually over approximately 1 min on ice, giving the source-selected 5% final DMSO branch.
4. Transfer immediately to the prechilled freezing container at -80 °C, then to liquid nitrogen within 2-3 days. Record container model, its specified cooling procedure and storage history. [S6 p.1]

TO COMPLETE - C01
Confirm derivative-specific banking and complete thaw/recovery SOPs, reagent formulations, cooling-container procedure and post-thaw qualification. The K562 note prefers IMDM for culture but names Jurkat medium during thawing; resolve that inconsistency. Its maintenance-density range lacks units and is not a transferable instruction.

Ready for the pairing assay
Release a qualified library with cell and construct identities, full reference sequences, expression and viability results, measured representation, culture/selection/banking history and defined acceptance criteria. Verify recovery after thaw before using the banked material. Attach the manifest to the run plan and preserve its version through analysis.

Source notes

Sources are the method authors' laboratory records, as transcribed in the supplied consolidation. The internal source files are not included in this download.

S1. Cell_x_Cell TCR_HLA-pep Pairing Assay V1. Author-supplied laboratory notes, p.1; construct architecture and input-library requirements. Statements and page references transcribed from the supplied source-checked synthesis dated September 11, 2026; underlying PDFs were not re-reviewed for this document.

S2. Protocol_TCRAFT_pooled vTim. Author-supplied notes, pp.2-3,12-13; optional upstream TCR construction and unresolved adaptation, as recorded in the supplied synthesis.

S3. PEP library cloning. Author-supplied notes, pp.1-4; HLA-peptide cloning, recipe gaps and defective-stock warnings, as recorded in the supplied synthesis.

S4. Generating Cell Lines for NxN Pairing Assay. Author-supplied notes, pp.1-2; packaging schedule, selection, representation and banking, as recorded in the supplied synthesis.

S5. Spinfection for Cell_x_Cell. Author-supplied notes, pp.1-2; dose examples, spinfection and conflicting selection timing, as recorded in the supplied synthesis.

S6. K562 Freeze Protocol. Author-supplied notes, pp.1-2; source-selected freezing branch and unresolved thaw medium, as recorded in the supplied synthesis.

S10. CZ PRIMERS AND GENEBLOCKS - CZP1-. Author-supplied registry, p.1; primer identities and pool-amplification conflict, as recorded in the supplied synthesis.

Protocol and resources